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From Excel Tariff Calculators to LLM-Powered Python Code

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Agenda: Your Journey Today

Roadmap

Time	Topic	Duration
09:30	Setup & Toolchain	30 Minutes
10:00	Analysing the Excel Calculator	15 Minutes
10:15	LLM Strategies: “Crafted” vs. “Industrial” Approaches	20 Minutes
10:35	Break	15 Minutes
10:50	Hands-on Porting: Testing, Maintainability & Walkthrough of Chat Protocols	100 Minutes
12:30	Wrap-up, QA Checklist & Next Steps	15 Minutes
12:45	End of the Session	–

The Right Development Environment for Your Project

VS Code: Modern, extensible IDE with excellent Python support and integrated terminal for rapid prototyping

GitHub Codespaces: Cloud-based development environment that provides instant access to preconfigured workspaces

Workflow Integration: Exemplary seamless connection between version control, testing, and deployment for professional development processes

Analysis of the Excel Calculator

Understand the structure and logic of the existing Excel tariff calculator

Formula Logic

Identification of formulas and analysis of dependencies between cells and cell ranges

VBA Modules

Analysis of existing functions and dependencies

Input/Output Mapping

Documentation of all input parameters and expected output values for validation purposes

Two Approaches for AI-Assisted Code Generation

Crafted Approach
Low entry barrier - ideal for rapid prototyping or one-time tasks

- Iterative, step-by-step development
- Copy formulas and VBA code into LLM
- Create Python code step by step with LLM
- Iterative error correction and testing at the end
- Ongoing control, but manual effort

Industrial Approach
Efficient and scalable, with transparent approach and full auditability

- Complete Excel-to-text extraction
- LLM pipeline: Automated generation with pytest gates
- Golden master comparisons and unit tests
- Human-in-the-Loop maximally efficient

Both approaches have their place – depending on goals, resources, and requirements.

Best Practices

Prompt Engineering Is Critical

Structure your requests precisely: Context → Task → Format → Constraints. The more detailed your prompt, the more precise the generated code.

Validation Through Test Cases

Create a comprehensive test dataset from your Excel model before porting. This allows you to systematically validate the Python implementation.

Documentation During Development

Record all AI interactions. These logs are valuable for traceability, quality assurance, and future optimizations.

Iterative Refactoring

The first generated code is rarely optimal. Plan time for incremental improvements: readability, performance, maintainability.

Hands-on: Live Porting with ChatGPT

100 minutes of intensive co-working – bring your questions!

01

Analysis & Preparation
Joint review of an example calculator and identification of porting goals

02

Code Generation
Live demonstration: From Excel sheet to first Python functions with ChatGPT

03

Testing & Debugging
Systematic comparison of results, error analysis, and iterative improvement

04

Lessons Learned & Optimization
Q&A after the exercise, refactoring for readability if needed, comparison with example repo – best practices in action

Maintainability & Quality Assurance

Unit Tests

Automated tests for every function guarantee that changes have no unexpected side effects. Use pytest for structured test suites.

Code Documentation

Docstrings, type hints, and inline comments make your code understandable for colleagues and your future self. Invest in clarity.

Version Control

Git enables traceable change history, branching for features, and secure team collaboration. Indispensable for professional development.

Continuous Integration

Automated build and test pipelines ensure that every code change meets quality standards before deployment.

Next Steps: Your Journey Continues

What You Learned Today

- Effective use of LLM tools in the process of porting an Excel calculator to Python code
- Best practices for porting, testing, and maintainability
- Practical experience with programming environments in VS Code or on GitHub Codespaces

Recommended Next Steps

- Apply what you learned to your own Excel tools
- Build a personal prompt library
- Network with the community for knowledge exchange

Questions? Discussion? Take the opportunity now!

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